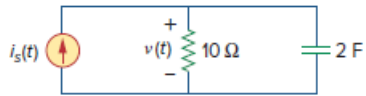


Problem

A sinusoidal current source of amplitude 4 A and frequency 2 kHz is applied to the circuit in Fig. 17.41. Use *PSpice* to find $v(t)$.

Figure 17.41:

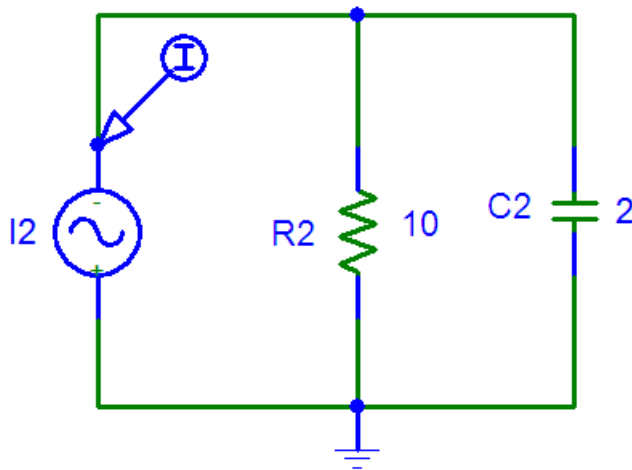


Step-by-step solution

Step 1 of 6

Using p spice to find the Fourier components consists of four steps.

Step:-1



The schematic for obtaining the Fourier coefficients

Step 2 of 6

Step:-2

We enter the attributes of the current source ISIN by double clicking on the voltage source and enter the values as

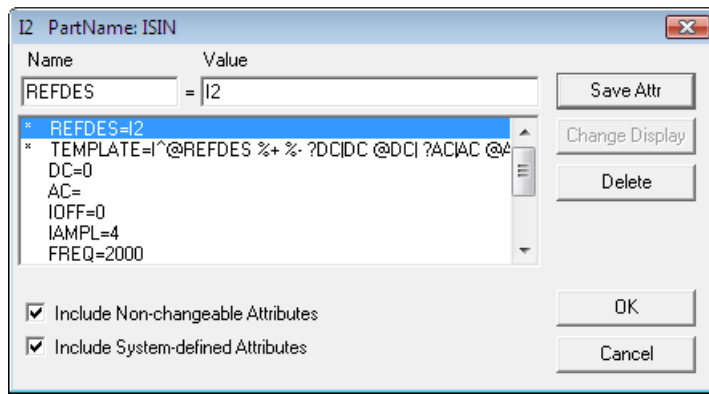
$$DC = 0$$

$$IOFF = 0$$

$$IAMPL = 4$$

$$FREQ = 2000$$

Step 3 of 6



Step 4 of 6

Step:-3

Given signal at a frequency of $f = 2\text{KHz}$

In the menu bar

Click **analysis** → **setup**

In the setup disable the Bias point and enable the **transient** and click on transient and enter the parameters as

Print step as 50μ

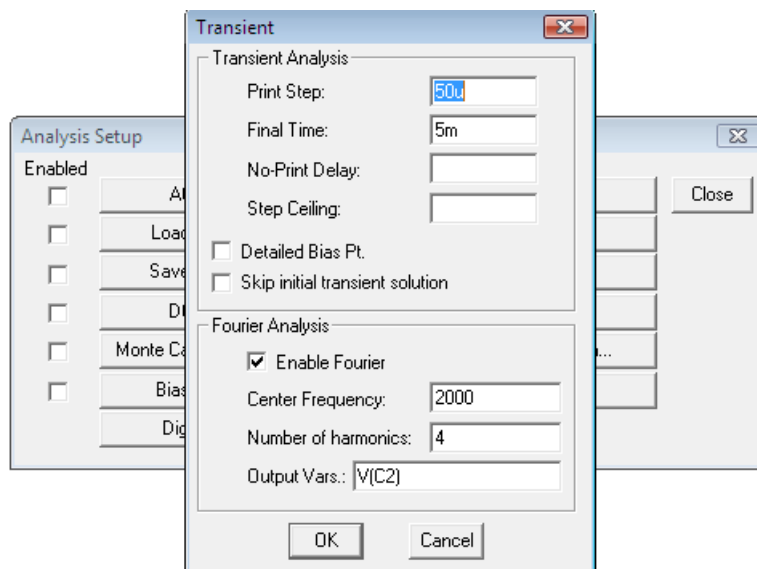
Final time T as 5m

Centre frequency as 2000

Number of harmonics as 4 and the

Output VARS as $V(C2)$

Step 5 of 6



Step:-4

Then save and click on simulate, when PSpice is run, the output file contains the result as

Step 6 of 6

FOURIER COMPONENTS OF TRANSIENT RESPONSE V(C_C2)

DC COMPONENT = -1.498130E-04

HARMONIC FREQUENCY FOURIER NORMALIZED PHASE NORMALIZED

NO (HZ) COMPONENT COMPONENT (DEG) PHASE (DEG)

1 2.000E+03 1.454E-04 1.000E+00 9.007E+01 0.000E+00

2 4.000E+03 1.796E-06 1.235E-02 9.713E+01 -8.301E+01

3 6.000E+03 1.370E-06 9.422E-03 9.523E+01 -1.750E+02

4 8.000E+03 9.760E-07 6.711E-03 8.209E+01 -2.782E+02